A Low-Cost LoRa-Based Environmental Monitoring System for Wildfire-Prone Areas in the Brazilian Savanna

Gustavo Blois
ICMC - USP
University of São Paulo
São Carlos, Brazil
gustavobloisrr@usp.br

Gabriel Cruz
ICMC - USP
University of São Paulo
São Carlos, Brazil
gabriel.melo.cruz@usp.br

Kalinka Branco
ICMC - USP
University of São Paulo
São Carlos, Brazil
kalinka@icmc.usp.br

***Abstract*—This study evaluates a low-cost, stationary LoRa environmental monitoring prototype for wildfire-danger assessment in the Brazilian Cerrado. Five field and indoor campaigns between April and July 2026 yielded 522 stored packets. Counter-based packet delivery during the recorded runs was 99.6–100%. A separate attempted deployment produced no cloud records. Two nodes 71 and 76 m from the gateway differed by 15.1 dB in mean received signal strength. One node operated for 221.4 h as its reported battery level fell from 69% to 2%. In 25 scheduled readings from the only functioning dry-season environmental node, relative humidity had a mean absolute difference of 17.1 percentage points against a nearby weather station. A one-day dew-point pattern is consistent with moisture cycling in the enclosure, and the node and station assigned different Ångström danger classes on that day. The pilot shows the value of tracking radio delivery and sensor validity alongside the readings.**

***Index Terms*—LoRa, environmental monitoring, wildfire danger, Brazilian Savanna, Cerrado, field deployment**

I. Introduction

The Brazilian Savanna, or Cerrado, hosts a rich diversity of endemic flora and fauna. It is a fire-adapted ecosystem, and natural burning plays an essential ecological role [1]. The same characteristic leaves it vulnerable to anthropogenic and illegal wildfires [2]. In 2024 the Atlantic Forest and the Amazon reached record burned area, and the Savanna remained among the most affected regions [3].

LoRa (Long Range) is a low-power wide-area radio technology used in environmental monitoring [4]. Here it connects fixed weather-sensing nodes to a gateway. The prototype records conditions relevant to fire danger, including air temperature and humidity. Fixed nodes of this kind form the ground segment of robotic wildfire monitoring: UAV-assisted LoRa surveillance [5] and a previously studied multi-hop network [6] both depend on stationary sensors whose readings are valid. This paper evaluates that ground segment in the field.

The pilot deployment took place in a Cerrado preservation area at the Federal University of São Carlos (UFSCar). The study examines received signal strength and packet delivery, the duration of one partial battery discharge, and differences between node readings, a nearby weather station, and public gridded products. The central question is whether radio delivery and environmental-data quality remain adequate together under field exposure.

II. Related Work

Wildfire-sensing systems have been evaluated under different conditions and for different outcomes. Vega-Rodríguez et al. tested a low-cost LoRa network for forest-fire detection [7]. Herring et al. evaluated a buried LoRaWAN node during a controlled bushfire, focusing on thermal survival and packet loss over short links [8]. Makdee et al. reported field communication and alarm tests for a solar-powered LoRa wildfire-warning system in Thailand [9]. Santos et al. combined LoRa sensors with a UAV and evaluated the resulting surveillance system [5]. A survey by Chan et al. discusses the measurement of energy use, latency, and detection accuracy in ground-based wildfire sensors [10]. An earlier Cerrado study examined multi-hop topology and single-hop range [6].

This study compares hourly enclosure measurements with a nearby station and examines their effect on Brazilian fire-danger indices. Differences in sensors, links, sites, and detection goals preclude a controlled numerical comparison with the cited systems. A same-site comparison with a calibrated instrument remains open.

III. Methodology

The proposed system consists of sensor nodes deployed in the monitored area and a central gateway that receives and forwards the collected data to a cloud server (Fig. 1).

A. Sensor Nodes

The sensor nodes deployed throughout the Brazilian Savanna preservation area are powered by 500 mAh LiPo batteries and built around an ESP32 LoRa Heltec V3 microcontroller [11], interfaced with a BME280 sensor [12] for temperature, relative humidity, and atmospheric pressure measurements. The datasheet specifies an absolute accuracy of $\pm 0.5\,^{\circ}\mathrm{C}$ at $25\,^{\circ}\mathrm{C}$ ($\pm 1.0\,^{\circ}\mathrm{C}$ from 0 to $65\,^{\circ}\mathrm{C}$), ± 3 percentage points of relative humidity from 20% to 80%, and ± 1.0 hPa of pressure.

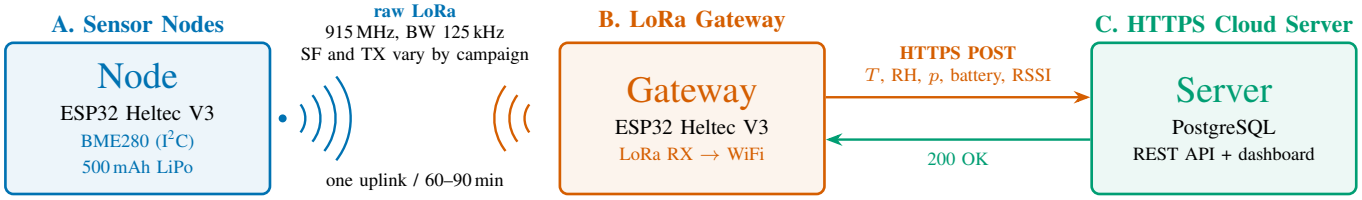


Fig. 1: Architecture of the custom point-to-point LoRa link and HTTPS backhaul. A radio acknowledgement precedes cloud ingestion. Table I lists the radio configurations.



(a) Sensor node inside the open enclosure.

(b) Node deployed on a tree.

Fig. 2: Sensor node hardware and deployment. (a) ESP32 Heltec V3, BME280, and 500 mAh LiPo battery housed in a waterproof enclosure modified with an air vent intended to permit ambient temperature and humidity measurement. (b) Node installed under vegetation cover in the study area.

Each node was housed in a nominally waterproof enclosure. F1 and F2 used sealed cases. F3 used a case with an air vent (Fig. 2). The BME280 measured air inside the enclosure, which could differ from ambient air. N4's BME280 was disconnected throughout F3. The field nodes did not carry cameras.

B. LoRa Gateway

The gateway uses the same Heltec platform and a custom raw-LoRa packet and acknowledgement protocol. A node sleeps between scheduled readings and retries up to three times if it receives no radio acknowledgement. The gateway acknowledges reception before forwarding the frame to the cloud over WiFi and HTTPS. The campaign firmware had no durable queue, so readings already acknowledged over radio could be lost during a backhaul outage.

C. HTTPS Cloud Server

The cloud server exposes an HTTPS REST API for ingesting and querying readings. The gateway supplies a pre-shared API credential in the request header. Readings are stored in PostgreSQL and displayed on a web dashboard. The packet carries a node identifier, sample counter, temperature, relative humidity, pressure, and estimated battery level. The gateway adds received signal strength (RSSI). Identifier bytes were not stable physical-node identities in these campaigns, so the dataset was reconciled by board and campaign.

D. Study Site and Reference Instrumentation

All field deployments took place in a Cerrado preservation area on the UFSCar campus. The gateway remained at $(-21.979322, -47.883435)$. INMET automatic weather station A711 is approximately 120 m from the gateway and publishes hourly precipitation, temperature, dew point, relative humidity, wind, and radiation [13]. The station provides a nearby reference for the node readings.

E. Campaign Design

Five campaigns ran between April and July 2026. Table I lists spreading factor (SF), transmit power (TX), reporting interval (Int.), enclosure, and node distance. Three field campaigns placed nodes 2 m above ground. Two indoor campaigns used the same boards a few metres from the gateway. Several settings, locations, enclosures, and seasonal conditions changed together. Comparisons between campaigns are therefore descriptive.

The F1 assignments of nodes to positions were reconstructed from the field notes and RSSI ordering because no

TABLE I: Campaign configurations. Distance is node to gateway. TX in dBm, Int. in minutes. All nodes used 500 mAh LiPo cells.

Exp	Window	SF	TX	Int.	Encl.	Nodes and distance
F1	28–30 Apr	7	14	90	sealed	N1 72.8 m; N2 94.4 m
I1	1–3 May	7	14	90	sealed	indoor, 4 m, through 1 wall
I2	15 May	10	16	90	sealed	indoor, 2 m, line of sight
F2	19–24 May	10	16	60	sealed	N3 150.8 m
F3	8–18 Jul	11	16	60	vented	N4 70.8 m; N5 75.8 m

position identifier was transmitted. The F2 and F3 positions were recorded directly.

The field nodes connected the BME280 to Heltec WiFi LoRa 32 V3 boards over I^{2}C on GPIO4 and GPIO3. The raw radio link used 915 MHz, 125 kHz bandwidth, and coding rate 4/5. Pipe-delimited frames carried node ID, readings, estimated battery level, an optional camera-battery field, and counter; the gateway added RSSI. Antenna gain was not recorded. The campaign firmware, raw packets, and scripts are available at `github.com/gmelodie/fire-lora-heltec`, commit `10160eb`. Later firmware revisions do not describe these campaigns.

F. Fire Danger Indices and Public Baselines

Two fire danger indices used in Brazilian forest management are computed from inputs sampled at 13:00 local time, where H is relative humidity in percent and T is air temperature in °C. The Monte Alegre Formula (FMA) was the first fire danger index developed in Brazil [14] and remains in operational use [15]. It accumulates the reciprocal of relative humidity over n consecutive days without rain,

$$\mathrm{FMA} = \sum_{i=1}^{n} \frac{100}{H_i}. \tag{1}$$

Being cumulative, (1) carries rainfall restrictions [15]: the accumulated value is reduced by 30%, 60% or 80% for daily rainfall of 2.5–4.9, 5.0–9.9 and 10.0–12.9 mm, and reset to zero above 12.9 mm. The Ångström index [16] is not cumulative and combines humidity with temperature,

$$B = \frac{H}{20} + \frac{27 - T}{10}, \tag{2}$$

and is read inversely, lower values indicating greater danger.

Each index value maps to a danger class. The FMA scale of Soares [14] defines *None* up to 1.0, *Low* from 1.1 to 3.0, *Medium* from 3.1 to 8.0, *High* from 8.1 to 20.0 and *Very high* above 20.0. The Ångström scale of Chandler et al. [17] defines *None* above 4.0 (fire unlikely), *Low* from 2.5 to 4.0 (conditions unfavourable), *Medium* from 2.0 to 2.5 (conditions favourable) and *High* below 2.0 (fire very likely).

The node has neither a rain gauge nor an anemometer. Node-based FMA uses A711 rainfall, while Ångström uses node data alone. Reference series use A711, ERA5 at its native 0.25° resolution through Open-Meteo [18], [19], and NASA POWER [20]. Processing excludes missing fields, a stuck

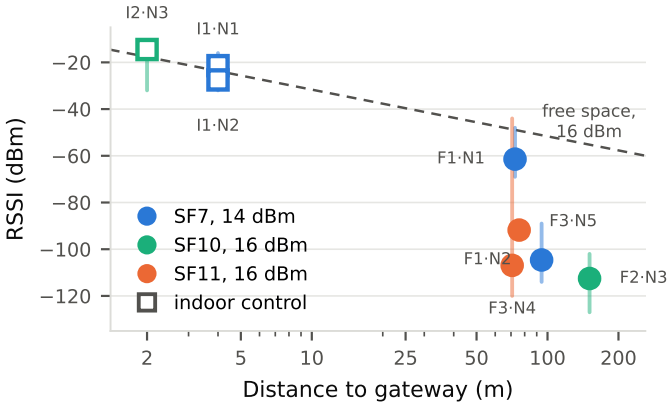


Fig. 3: Received signal strength (RSSI) by run. Setup bursts are excluded from N5. Indoor controls are qualitative because of receiver compression. The dashed line shows the free-space reference of (3).

100 % humidity series, and values outside $-10 < T < 55$ °C, $0 < H < 100$ %, and $890 < p < 950$ hPa. An index requires a valid reading within 30 minutes of 13:00 local time and, for FMA, known daily rainfall. Each short node FMA segment starts from the previous day's A711 accumulator and uses A711 rainfall; the station and gridded series provide the longer comparison.

For environmental agreement, each scheduled N3 reading in F2 and N5 reading in F3 is paired with the nearest A711 hourly record. The reported metrics are signed bias (node minus station), mean absolute error (MAE), root mean square error (RMSE), and Pearson correlation. Pressure is compared without altitude correction. Setup bursts with repeated counter zero are excluded. N3 gives 110 eligible readings over 4.6 days, and N5 gives 25 readings over about 25 h. Only one day supplies an N5 reading at the index hour. The 95% confidence intervals come from a circular moving-block bootstrap [21] with 10 000 resamples. A block holds 24 consecutive hours, so each resample keeps the daily cycle of the enclosure error. For N5 the block is 6 h, so that each resample draws at least four blocks; the N5 intervals therefore understate the uncertainty of a one-day record.

IV. Results

The five selected campaign windows contain 522 stored packets (Table III). This includes some N5 setup bursts, which are excluded from environmental analysis.

A. Link Quality and Delivery

Fig. 3 plots RSSI against distance. The dashed reference is the Friis free-space model [22],

$$P_r = P_t + G_t + G_r - 20\log_{10} d - 20\log_{10} f - 32.44, \tag{3}$$

with P in dBm, gains G in dBi, d in kilometres and f in megahertz, plotted for 16 dBm at 915 MHz with an assumed 2 dBi at each end. N4 at 70.8 m averaged −106.9 dBm against −91.8 dBm for N5's 25 scheduled transmissions at 75.8 m.

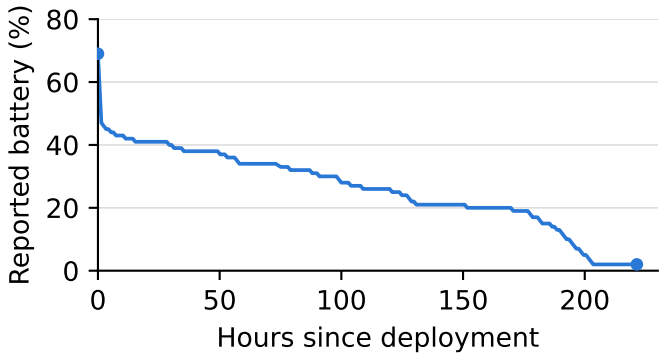


Fig. 4: Reported battery level during F3. N4's voltage-derived estimate fell from 69% to 2% over 221.4 h. The percentage was not calibrated against discharged capacity.

Free-space loss changes by only 0.6 dB across that distance difference. Canopy, orientation, mounting, and multipath may contribute to the 15.1 dB RSSI gap, although their separate effects cannot be determined here.

Counter continuity in the recorded scheduled stretches indicated 99.6–100% post-retry delivery to the gateway. N4 missed one counter in 223 recorded packets across 9.2 days. The weakest observed frame, from F2, had RSSI −127 dBm. N5's counter restarted during setup, so its delivery ratio is omitted. On 3 July, a deployment attempt produced no stored readings. Node-side logs and gateway heartbeats are unavailable for that attempt, so the point of failure and its duration are unknown. The packet-delivery percentages describe only the radio link during recorded runs.

B. Power Consumption

N4 remained active for 221.4 h while its voltage-derived battery estimate fell from 69% to 2% (Fig. 4). This gives 9.2 days of observed operation from a partly charged 500 mAh cell under the F3 duty cycle. The percentage comes from a nonlinear voltage lookup. Current was not logged, and the run did not start with a full charge. N4 also had no functioning BME280. The firmware used timed deep sleep between transmissions, but its effect on energy use was not isolated experimentally.

The F1 nodes drained faster: N1's estimate fell from 91 % to 0 % in about two days. This is likely due to the fact that the F1 firmware did not yet use deep sleep. The campaigns therefore do not share a single power profile.

C. Environmental Agreement

Table II summarizes the N3 and N5 comparisons with A711. Over the 4.6 days of F2, N3 read 0.81 °C cooler than the station, outside the ±0.5 °C datasheet accuracy, and 6.80 percentage points wetter, with an interval of 1.52 to 11.72 points. Its humidity correlated weakly with the station ($r = 0.22$) because the air in the sealed case grew wetter every day, to a daily mean of 97.3 %. Pressure read 1.66 hPa low at N3 and 2.47 hPa low at N5, beyond the ±1.0 hPa specification; at roughly 0.12 hPa per metre, a height difference of 14 to 21 m from the station barometer gives the same offsets. Over N5's short window, temperature and pressure followed the station closely. Relative humidity varied more, with an MAE of 17.06 percentage points and a correlation of 0.59. The station is roughly 120 m from the gateway, so these differences combine instrument, enclosure, and local microclimate effects.

TABLE II: N3 (F2) and N5 (F3) against A711, paired by the nearest hourly record. Bias is node minus station. RH is relative humidity, with differences in percentage points. Correlation is Pearson's r. Brackets give 95% block bootstrap confidence intervals, with 24 h blocks for N3 and 6 h blocks for N5.

Variable	Bias	MAE	RMSE	r
F2: N3, 19–24 May, 110 readings				
Temp. (°C)	−0.81 [−1.01, −0.63]	0.90 [0.70, 1.08]	1.02 [0.82, 1.19]	0.98 [0.97, 0.99]
RH (pp)	+6.80 [+1.52, +11.72]	8.55 [5.24, 11.87]	11.62 [7.20, 14.99]	0.22 [0.04, 0.89]
Press. (hPa)	−1.66 [−1.79, −1.51]	1.66 [1.51, 1.79]	1.68 [1.54, 1.81]	0.97 [0.95, 0.98]
F3: N5, 8–9 July, 25 readings				
Temp. (°C)	−0.36 [−0.98, +0.32]	1.18 [0.57, 2.02]	2.00 [0.67, 3.17]	0.94 [0.84, 0.99]
RH (pp)	−4.42 [−18.27, +10.60]	17.06 [13.69, 20.43]	18.52 [15.64, 21.01]	0.59 [−0.17, 0.92]
Press. (hPa)	−2.47 [−2.81, −2.08]	2.47 [2.08, 2.81]	2.51 [2.12, 2.81]	0.97 [0.90, 0.99]

The dew point calculated from N5 was about 4–5 °C below A711 before dawn on 9 July. At 09:44 local it was 14.0 °C above A711, when N5 also read 8.0 °C warmer and 27.2 percentage points wetter. Moisture evaporating inside the sun-warmed enclosure is one explanation for this reversal. A sensor outside the case and a co-located reference would help distinguish it from local air differences or sensor error. At 14:13 local, N5 recorded 58.0% relative humidity against 34% at A711. This 24-point difference changes the humidity term used by both fire-danger indices.

D. Fire Danger Indices Against Public Data

Fig. 5 shows FMA from the station and public products across the available record. On 18 May A711 recorded 13.8 mm of rain, which resets FMA to zero. ERA5 recorded 5.5 mm for its grid cell and yielded 5.9, in the *Medium* class, against *None* at A711. On 22 May the values were 9.3 and 4.6. NASA POWER also differed from A711, reaching an FMA of 94 against 55. Its 13:00 temperature was 2.5–8 °C above the station during the comparison. The product's lower grid-cell elevation may contribute, along with differences in spatial averaging and weather representation.

Fig. 6 compares the Ångström index on days with an eligible node reading. On 9 July A711 yielded 2.31 (*Medium* danger), while N5 yielded 3.32 (*Low*). NASA POWER yielded 1.64 (*High*) and ERA5 2.61 (*Low*). A station-based dew-point adjustment puts N5 near 2.30 in the figure, but it uses A711 to set the adjustment. The observed node and station

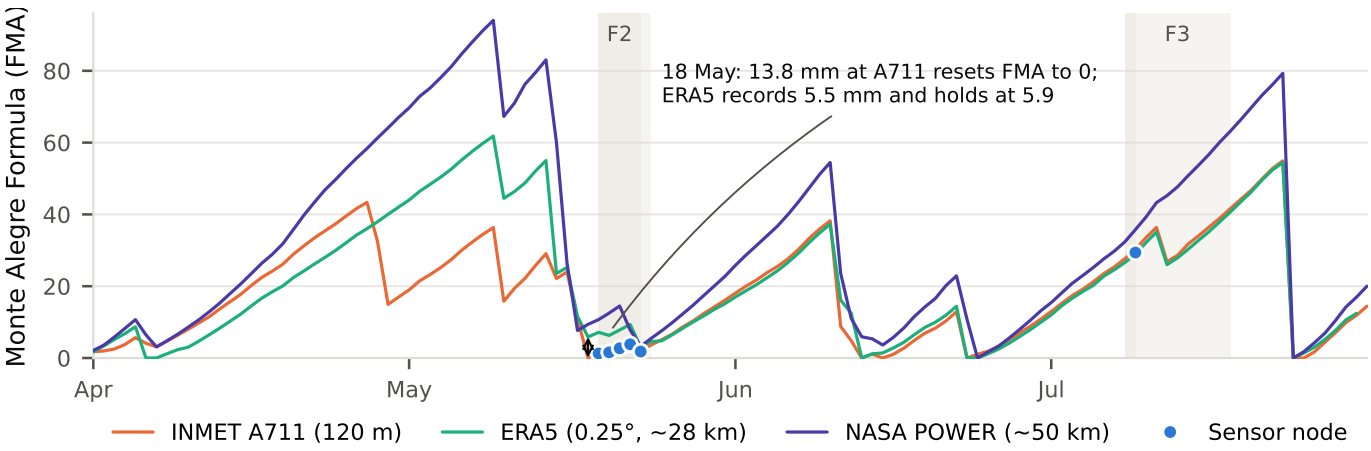


Fig. 5: Monte Alegre Formula, 1 April to 31 July 2026, from A711, ERA5 and NASA POWER, with the six days a node reported at 13:00 local. Light columns mark the F2 and F3 campaigns, darker columns the sub-periods with a working humidity sensor.

classifications disagree on this day. Over the six days with an eligible node reading at 13:00 local, the unadjusted node and A711 assigned the same FMA class on five days and the same Ångström class on three. FMA differed on 21 May (*Low* at the node, *Medium* at A711). Ångström differed on 21 and 22 May, where the wetter N3 read *None* against *Low*, and on 9 July. More index-hour samples are needed to assess classification accuracy.

The node-based FMA series is less independent because it uses A711 rainfall and starts from the station's prior accumulation. Its close agreement with A711 during F2 should be read in that context. The Ångström comparison uses only the node's temperature and humidity, but the available N5 index sample covers a single dry-season day.

E. Limitations

Short campaigns and few working probes constrain the comparison.

Enclosure water ingress. Moisture intrusion affected two campaigns. N2 reported 100.00 % humidity in all 33 outdoor packets, consistent with sensor damage. N3 lost contact at 68 % battery after internal humidity rose to a daily mean of 97.3 %. The vented N5 case also showed a strong daily humidity discrepancy, of unmeasured cause.

Receiver saturation. Above about −25 dBm, receiver gain compression makes reported RSSI underestimate signal strength. Indoor controls (I1, I2) averaged −14.5 to −27.5 dBm and serve only as qualitative references. Field measurements were all below −60 dBm.

Temporal coverage. The node dataset has five eligible index-hour days in May and one in dry-season July. N5's 25 readings cover only about a day. Radio settings, position, enclosure, and season also changed together, preventing causal attribution or a seasonal accuracy estimate.

TABLE III: Stored packets in selected windows. PDR is counter-based and post-retry where the counter is continuous. Sensor gives valid scheduled environmental readings. N5 has 36 stored packets, including setup bursts, and 25 eligible hourly samples. Battery percentages are voltage estimates.

Exp	Node	Pkts	PDR	RSSI	Batt.	Sensor
F1	N1	33	100 %	−61.4	91–0	0/33
F1	N2	33	100 %	−104.6	98–45	0/33
I1	N1	37	100 %	−21.4	57–52	0/37
I1	N2	37	100 %	−27.5	34–0	0/37
I2	N3	10	100 %	−14.5	93–89	10/10
F2	N3	113	100 %	−112.3	74–68	110/113
F3	N4	223	99.6 %	−106.9	69–2	0/223
F3	N5	36	n/a	−91.8	16–7	25/25

V. Conclusion

The Cerrado pilot showed high post-retry delivery in recorded runs, although the 3 July attempt produced no cloud records. N4 operated for 9.2 days from a partially charged cell.

N5 showed a 17.1-point humidity MAE against A711; on one dry-season day, the Ångström class was Medium at the station and Low at the node. Across six index days, node and station agreed on FMA class five times and Ångström class three times. Rainfall differences also caused FMA to diverge between A711 and ERA5. Longer, co-located measurements are needed before operational use.

Future work will compare sealed and vented cases at one site, measure a full discharge, and log node and gateway uptime. A mobile robot or UAV carrying a calibrated sensor could map microclimates and guide node placement; its navigation and sensing would require separate evaluation. Sampling across Cerrado vegetation formations [23] would test how well fixed nodes represent wider areas.

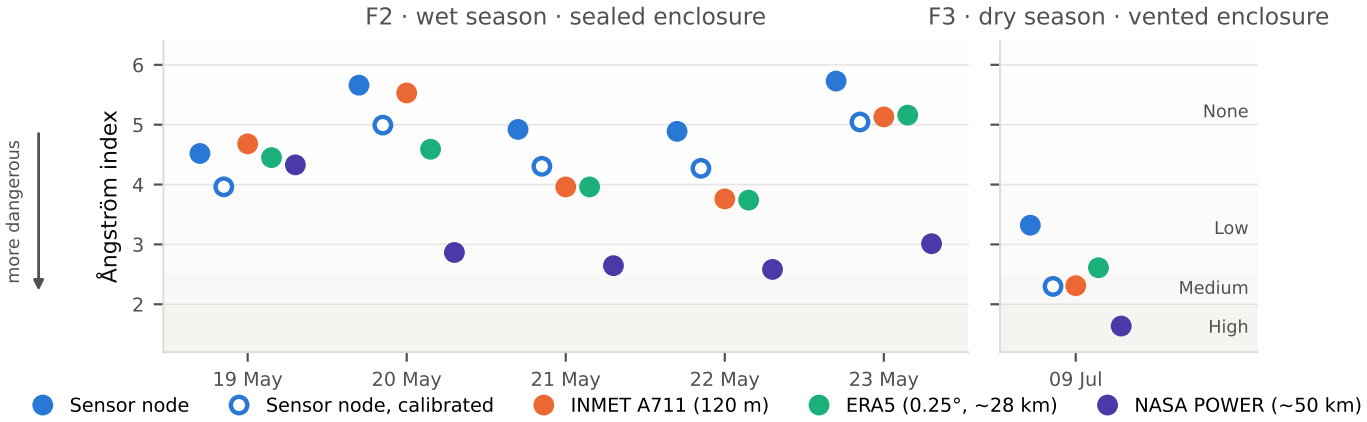


Fig. 6: Ångström index by source on days with a valid node sample. Lower values indicate greater danger. Hollow markers show the node series after an A711-based dew-point adjustment, for illustration. Shaded bands mark the classes of Chandler et al. [17]. On 9 July the nearby station is in the Medium class and the unadjusted N5 is in the Low class.

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